

Task:-6

PL/SQL Procedures, Functions, Loops

Aim:-

To implement PL/SQL Procedures, Functions and loops on Number theory and business scenarios.

Procedure:-

PL/SQL is a combination of SQL along with the procedural features of programming languages. It was developed by Oracle Corporation in the early 90s to enhance the capabilities of SQL. PL/SQL is one of three key programming languages embedded in the Oracle Database, along with SQL itself and Java.

Sr/o

Sections & Description

1 Declaration

This section starts with the keyword DECLARE. It is an optional section and defines all variable, cursor, subprograms (and other elements to be used in the program.

2 Executable commands

This section is enclosed between the keywords BEGIN and END and it is a mandatory section. It consists of the executable PL/SQL statements of the program. It should have at least one executable line of code, which may be just a null command to indicate that nothing should be executed

3- Exception handling

This section starts with the keyword EXCEPTION. This optional section contains exception(s) with that handle errors in the program.

Simple program to print a sentence:

Syntax:

```
DECLARE  
  <declarations section>  
BEGIN  
  <executable commands>  
EXCEPTION  
  <exception handling>
```

Program:

```
DECLARE  
message varchar2(200) := 'working code';  
BEGIN  
  dbms_output.put_line(message);  
END;
```

Static input:

```
SQL> set serveroutput on  
SQL> declare
```

```
2 x number(5);  
3 y number(5);  
4 z number(9);
```

```
5 begin
```

```
6 x:=10;
```

```
7 y:=12;
```

```
8 z:=x+y;
```

```
9 dbms_output.put_line('sum is' || z);
```

```
10 end;
```

```
11 /
```

```
sum is 22
```

PL/SQL procedure successfully completed.

dynamic input:

```
SQL> declare
2 var1 integer;
3 var2 integer;
4 var3 integer;
5 begin
6 var1 := 4;
7 var2 := 4;
8 var3 := var1 + var2;
9 dbms_output.put_line (var3);
10 end;
11 /

Enter value for var1: 20
old 6: var1:=4 var1;
new 6: var1:=20;

Enter value for var2: 30
old 7: var2:=4 var2;
new 7: var2:=30;

SQL
```

PL/SQL procedure successfully completed.

```
DECLARE
hid number(30):=100;

BEGIN
IF (hid=10) THEN
dbms_output.put_line ('Value of hid is 10');
ELSEIF (hid=20) THEN
dbms_output.put_line ('Value of hid is 20');
ELSEIF (hid=30) THEN
dbms_output.put_line ('Value of hid is 30');
ELSE
dbms_output.put_line ('None of the values is matching');
END IF;
dbms_output.put_line ('Exact value of hid is: '||hid);
END;
```


None of the values is matching
Exact value of hid is: 100

PLSQL procedure successfully completed

DECLARE

hid number(10);

oid number(10);

BEGIN

<< outer loop >>

FOR hid IN 1..3 LOOP

<< inner-loop >>

FOR oid IN 1..3 LOOP

dbms_output.put_line ('hid is: '|| hid||' and oid is: '|| oid);

END LOOP inner-loop;

END LOOP output-loop;

END;

/

hid is: 1 and oid is: 1

hid is: 1 and oid is: 2

hid is: 1 and oid is: 3

hid is: 2 and oid is: 1

hid is: 2 and oid is: 2

hid is: 2 and oid is: 3

hid is: 3 and oid is: 1

hid is: 3 and oid is: 2

hid is: 3 and oid is: 3

PLSQL procedure successfully completed

Sample program as only procedure:

SQL> create or replace procedure c8 in 8mahom

2 << cid in number, c-name in varchar2

3 is

4 begin

5 dbms_output.put_line ('cid: '|| cid);

6 dbms_output.put_line ('name: '|| c-name);

7 end;

8

Procedure created.

```
SQL> exec cs information 101, 'raam's';  
PL/SQL procedure successfully completed.  
SQL> set serveroutput on;  
SQL> exec cs information 101, 'raam's';  
101101
```

NAME: raam

PL/SQL procedure successfully completed

Sample program as only function

SQL> create or replace function cs information

(ch_id in number, l_name in varchar2)

return varchar2

is

begin

if c_id > 200 then

return ('no booking available');

else

return ('booking open');

end if;

end;

function created

SQL> declare

l_msg varchar2(200);

begin

l_msg := cs information 2 (102, 'raam's');

dbms_output.put_line (l_msg);

end;

7 /

Vehicle available

SQL > declare
2 msg varchar(200);

3 begin

4 msg := confirmation(206, 'ram');

5 dbms_output.put_line(msg);

6 end;

7 1
no vehicle available

SQL procedure successfully completed.

RESULT - Thus, the PL/SQL Procedure, Function, Loop system was successfully executed and completed.

TASK-6: PL/SQL Procedures, Functions, Loops

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```
SQL> connect
Enter user-name: system
Enter password:
ERROR
ORA-01017: invalid username/password; logon denied
```

```
SQL> connect
Enter user-name: system
Enter password:
Connected
SQL> set serveroutput on
SQL> declare
2 x number(5);
3 y number(5);
4 z number(9);
5 begin
6 x:=10;
7 y:=12;
8 z:=x+y;
9 dbms_output.put_line('sum is'|| z);
10 end;
11 /
sum is22
```

PL/SQL procedure successfully completed.

```
SQL> declare
2 var1 integer;
3 var2 integer;
4 var3 integer;
5 begin
6 var1=&var1;
7 var2=&var2;
8 var3=var1+var2;
9 dbms_output.put_line(var3);
10 end;
11 /
```


Enter value for var1: 20
old 6 var1 = &var1;
new 6 var1 = 20;
Enter value for var2: 30
old 7 var2 = &var2;
new 7 var2 = 30;
50

PL/SQL procedure successfully completed.

```
SQL> create or replace procedure csinformation
2 (c_id in number, c_name in varchar2)
3 is
4 begin
5 dbms_output.put_line('ID: ' || c_id);
6 dbms_output.put_line('name: ' || c_name);
7 end;
8 /
```

Procedure created.

```
SQL> exec csinformation(101, 'raam');
ID 101
name raam
```

PL/SQL procedure successfully completed.

```
SQL> set serveroutput on;
SQL> exec csinformation(101, 'raam');
ID 101
name raam
```

PL/SQL procedure successfully completed.

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EX NO.	668
PERFORMANCE (5)	5
RESULT AND ANALYSIS (5)	19
VIVA VOCE (5)	4
RECORD (5)	19
TOTAL (20)	48
DATE	28/08/23